

Climate-driven shifts in suitable areas of *Alternaria* leaf blotch (*Alternaria mali* Roberts) on apples: Projections and uncertainty analysis in China

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ABSTRACT

Apple production in China faces significant threats from *Alternaria* leaf blotch (ALB), a disease potentially exacerbated by climate change through shifts in its distribution and severity. However, the impacts of future climate change on ALB distribution remain insufficiently explored. We collected ALB occurrence data from orchard surveys, public databases, and the literature. Using five species distribution models (SDMs), we examined the relationship between environmental variables and ALB occurrence, and assessed its potential distribution under different climate change scenarios. The analysis used five Global Climate Models (GCMs) from the CMIP6 dataset, with a baseline period (1970–2000) and projections for the 2030s, 2050s, 2070s, and 2090s, based on four shared socioeconomic pathways (SSP126, SSP245, SSP370, and SSP585). The SDMs showed high reliability, with average values for the area under the receiver operating characteristic curve exceeding 0.96 and the true skill statistic exceeding 0.86. During the baseline period, ALB-suitable areas were primarily concentrated in the Bohai Bay, Loess Plateau, and Old Course of the Yellow River apple-planting regions. By the 2090s, under the SSP126, these areas were projected to decrease by 8.89 %. In contrast, under the SSP245, SSP370, and SSP585 scenarios, they were expected to increase by 4.89 %, 21.30 %, and 23.22 %, respectively, with a northwestward shift of 137 to 263 kms and an elevation increase of 288 to 680 m. Additionally, our findings indicated that GCMs contribute 42.2 % of the uncertainty in predictions, while SDMs and scenarios contribute 31.5 % and 8.28 %, respectively. This research highlights the importance of using multiple models and scenarios to enhance the accuracy of disease distribution predictions under changing climatic conditions. By identifying potential future hotspots and suitable areas of ALB, the study provides critical insights for safeguarding apple production in China against the impacts of climate change.

1. Introduction

Alternaria leaf blotch (ALB) of apple, caused by the pathogen *Alternaria mali* Roberts (AMR) is the most significant foliar disease in China's apple-planting regions (Wang et al., 2018b). It poses a serious threat to the apple industry, affecting both the economic viability of apple production and the sustainability of farming practices (Hu et al., 2016; Pan et al., 2012; Tian et al., 2024). ALB primarily affects leaves, leading to defoliation, reduced tree vigor, and damage to new shoots and fruits, which in turn degrades fruit quality (Gs et al., 2020; Hu et al.,

2005). Since its discovery in the Bohai Bay apple-planting region in the 1970s, ALB has been reported annually in most of China's apple-planting regions and caused significant losses (Hu et al., 2016). For instance, a severe outbreak in Tianshui city in 2022 reduced apple production by approximately 15 %, resulting in economic losses of 137.67 million USD (Wang, 2024). These impacts underscore the vulnerability of current apple production systems to external factors, including climate variability.

To manage ALB, apple growers in China typically employ both preventive and curative measures, with heavy reliance on fungicides (Li

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et al., 2013; Wang et al., 2018b). For instance, in Yulin, Shaanxi province, farmers begin spraying fungicides in May, applying them every 10–15 days. If persistent rain occurs, additional systemic fungicide applications are necessary once the rain stops (Ma et al., 2019). However, these experience-based management practices are increasingly challenged by climate change. Climate change is expected to alter key climatic factors—such as temperature, precipitation, and humidity—that directly affect the occurrence, spread, and severity of ALB (Hu et al., 2005, 2006; Pan et al., 2012; Wang et al., 2018b). They can lead to shifts in pathogen distribution, expanding the spread of crop diseases into new areas (Chakraborty and Newton, 2011; Chaloner et al., 2021; Raza and Bebbler, 2022; Singh et al., 2023). Recent studies have demonstrated a significant increase in the occurrence of crop pests and diseases in China since 1970, underscoring the urgency of understanding climate-disease interactions (Wang et al., 2022). Gaining insight into the relationship between ALB and climate change is crucial for developing effective adaptive management strategies. Farmers will need to adjust their fungicide application schedules and strategies to align with the new environmental conditions (Rossi et al., 2010). This may include altering the timing, frequency, and types of fungicides used to maintain control over ALB under climate change (Singh et al., 2023).

Recently, species distribution models (SDMs) have become invaluable tools for predicting the distribution of crop diseases and pests in relation to environmental variables (Bai et al., 2022; Ikeda and Osawa, 2020; Narouei-Khandan et al., 2016; Zhao et al., 2022). These models integrate factors such as temperature, precipitation, and humidity, providing critical insights into how climate change might influence the spread of crop diseases like ALB (Batista et al., 2023; Xu et al., 2020). By modeling the relationship between environmental conditions and pathogen occurrence, SDMs can forecast potential shifts in disease distribution under climate change. SDMs have been widely used to identify suitable areas for various crop diseases, offering valuable information for targeted disease management (Godefroid, 2023; Xu et al., 2020; Zhao et al., 2022). However, the potential impacts of climate change on the future distribution of ALB, particularly across China's diverse apple-planting regions, remain insufficiently explored.

It is essential to understand the impact of climate change on ALB suitable areas (Pan et al., 2012). Economically, increased disease prevalence can lead to higher management costs, reduced profitability, and adverse effects on smallholder farmers (Zaller et al., 2023). Anticipating shifts in ALB distribution allows for proactive management strategies, efficient resource allocation, and targeted interventions, thereby minimizing disease spread and reducing environmental impacts from excessive fungicide use. Additionally, insights from SDMs inform policy-making and agricultural planning, supporting affected regions and promoting research into resistant varieties and sustainable farming practices (Bai et al., 2022; Narouei-Khandan et al., 2016). Ultimately, understanding ALB's response to climate variables helps build resilient agricultural systems, securing farmers' livelihoods and ensuring the sustainability of apple cultivation in the face of global climate change.

In this study, we utilize five SDMs to establish the relationships between environmental variables and the occurrence of ALB, as well as to assess the relative importance of these variables. Employing an ensemble modeling approach, we project ALB-suitable areas under four climate change scenarios of the Shared Socio-economic Pathways (SSP) (SSP126, SSP245, SSP370, and SSP585) across four future time periods (2030s, 2050s, 2070s, and 2090s). We evaluate the impacts of climate change on the spatial and temporal shifts conducive to ALB. Additionally, we examine how different SDMs, general circulation models (GCMs), and climate scenarios contribute to uncertainties in our predictions, exploring their influence on ALB suitability assessments over various temporal scales. This comprehensive analysis enhances our understanding of ALB distribution dynamics and informs strategies for managing its spread under changing climatic conditions.

2. Materials and methods

2.1. Overall workflow

The overall workflow of this study is shown in Fig. 1. Initially, 1465 occurrence records of ALB, along with 19 bioclimatic and 4 topographic variables, were collected and preprocessed. This process involved generating absence data for ALB and reducing multicollinearity among the environmental variables. Subsequently, five SDMs, including Boosted Regression Trees (BRT), Multiple Adaptive Regression Splines (MARS), Maximal Entropy (MaxEnt), Random Forest (RF), and Support Vector Machines (SVM) were calibrated using a 70 %:30 % training-to-testing dataset split. The models were then evaluated using the area under the receiver operating characteristic curve (AUC) and the true skill statistic (TSS) metrics. Finally, the calibrated models were applied to baseline and future climate data to project shifts in suitable areas of ALB, followed by an uncertainty analysis of the predictions.

2.2. Study area

Apple cultivation in China spans 23 provinces and regions across the mainland China. Based on the China Apple Industry Development Zoning (2017) and related documents from the Ministry of Agriculture and Rural Affairs, Zhang (2021) categorized China's apple-producing provinces into five main regions: the Bohai Bay (including Shandong, Hebei, Liaoning, Tianjin, and Beijing), the Loess Plateau (including Shaanxi, Gansu, Shanxi, Ningxia, and Qinghai), the Old Course of the Yellow River (including Henan, Anhui, and Jiangsu), the Southwest Cold Highland area (including Sichuan, Yunnan, Guizhou, and Tibet), and the Specialty region (including Inner Mongolia, Heilongjiang, Jilin, and Xinjiang). These regions exhibit significant variability in elevation and climate conditions, with average elevations ranging from 146 to 3526 m, annual temperatures from 2.93 °C to 14.9 °C, and annual precipitation from 288 mm to 1011 mm (Fig. 2b, c, and d).

2.3. Presence and absence data of ALB

Presence-absence data are essential for training SDMs. The workflow for ALB presence data collection, processing, and integration is illustrated in Fig. 3.

For ALB presence data, records were compiled from orchard surveys, public databases, and published literature.

(1) Orchard survey

Between 2021 and 2022, we conducted field surveys using the “NongQingZhuShou” mobile application on the WeChat platform to collect data on ALB occurrence. Technicians recorded geographic coordinates and marked the ALB status as “occurred” when significant symptoms were observed. An example of an ALB survey conducted in an apple orchard in Wuqian Town, Yangling, Shaanxi Province is shown in Fig. S1.

(2) Public database

The second data source was the National Apple Big Data Public Data Channel (<https://appledata-channel.agri.cn/>), which provides information on apple pests and diseases at the county and district levels from 2011 to 2019. For this study, we specifically focused on the ALB occurrence data and carried out following preprocessing steps to ensure data accuracy and consistency. Firstly, we excluded provincial and municipal-level data, retaining only county-level records to ensure spatial consistency. To address the potential overestimation of ALB occurrence in areas with small reported sizes, we plotted histograms of ALB occurrence areas for each district (Fig. S2) and applied a threshold at the 75th percentile (2000 ha). Areas smaller than this threshold were

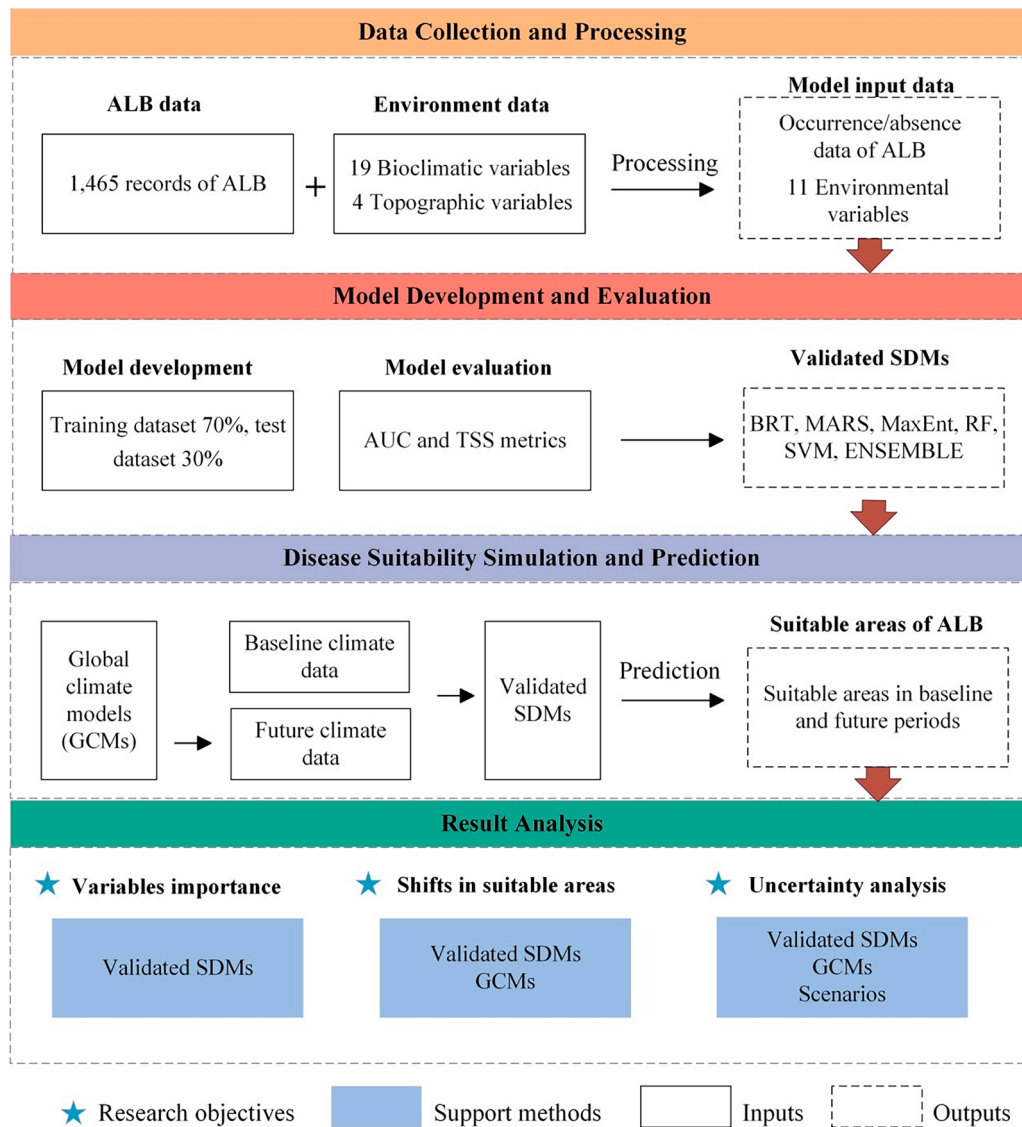


Fig. 1. The overall workflow of this study. Abbreviated interpretation: ALB: Alternaria leaf blotch; BRT: Boosted Regression Trees; MARS: Multiple Adaptive Regression Splines; MaxEnt: Maximal Entropy; RF: Random Forest; SVM: Support Vector Machines; AUC: the area under the receiver operating characteristic curve; TSS: the true skill statistic; GCMs: Global Climate Models.

set to 0, while the larger areas were retained. Secondly, using the corrected data, we calculated the average ALB occurrence area for each county over the period from 2011 to 2019. If the average area was greater than 0, we considered that ALB had occurred in that county during this time period. Finally, we utilized the Amap API (<https://lbs.amap.com/>) to retrieve the geographic coordinates of each county and standardized the data into a unified format, which included both the geographic coordinates and the “occurred” status for ALB.

(3) Historical literature

We collected ALB occurrence data from published literature by searching for the term “Apple” AND “Alternaria blotch” in Web of Science (<https://www.webofknowledge.com>) and CNKI (<https://www.cnki.net>). Our search yielded 10 peer-reviewed articles in Web of Science, along with 71 peer-reviewed articles and two book chapters in CNKI. Most articles focused on field investigations of ALB conducted between 1970 and 2000 in China. We extracted relevant data from the literature, using specific coordinates provided in some studies (e.g., Pan et al., 2012). For other sources that only recorded locations, we also used

the Amap API (<https://lbs.amap.com/>) to obtain geographic coordinates. To confirm the occurrence of ALB, we followed the judgment of the original authors. All data were then standardized into a unified format that included geographic coordinates and the “occurred” status for ALB.

After processing, a total of 1465 ALB occurrence records were collected (Fig. S3). All record points were harmonized using climate grids with a resolution of 4.3 km × 4.3 km based on bioclimatic variables from WorldClim (<https://www.worldclim.org/>). These grids represent 30-year (1970–2000) historical averages, providing a consistent temporal framework for analysis (Fig. 3). The accuracy of the three data sources varies: Orchard survey data is the most accurate due to direct field observations by trained investigators, followed by public database data, which is moderately accurate but limited by reporting standards and coverage. Historical literature data exhibits the highest variability in accuracy, depending on the quality of individual studies. However, after rigorous quality control, we treated all three data sources as equally reliable in the study. Each record represents an occurrence of ALB at a specific location. For temporal aggregation, a grid cell is designated as positive if at least one ALB case was observed there. Both

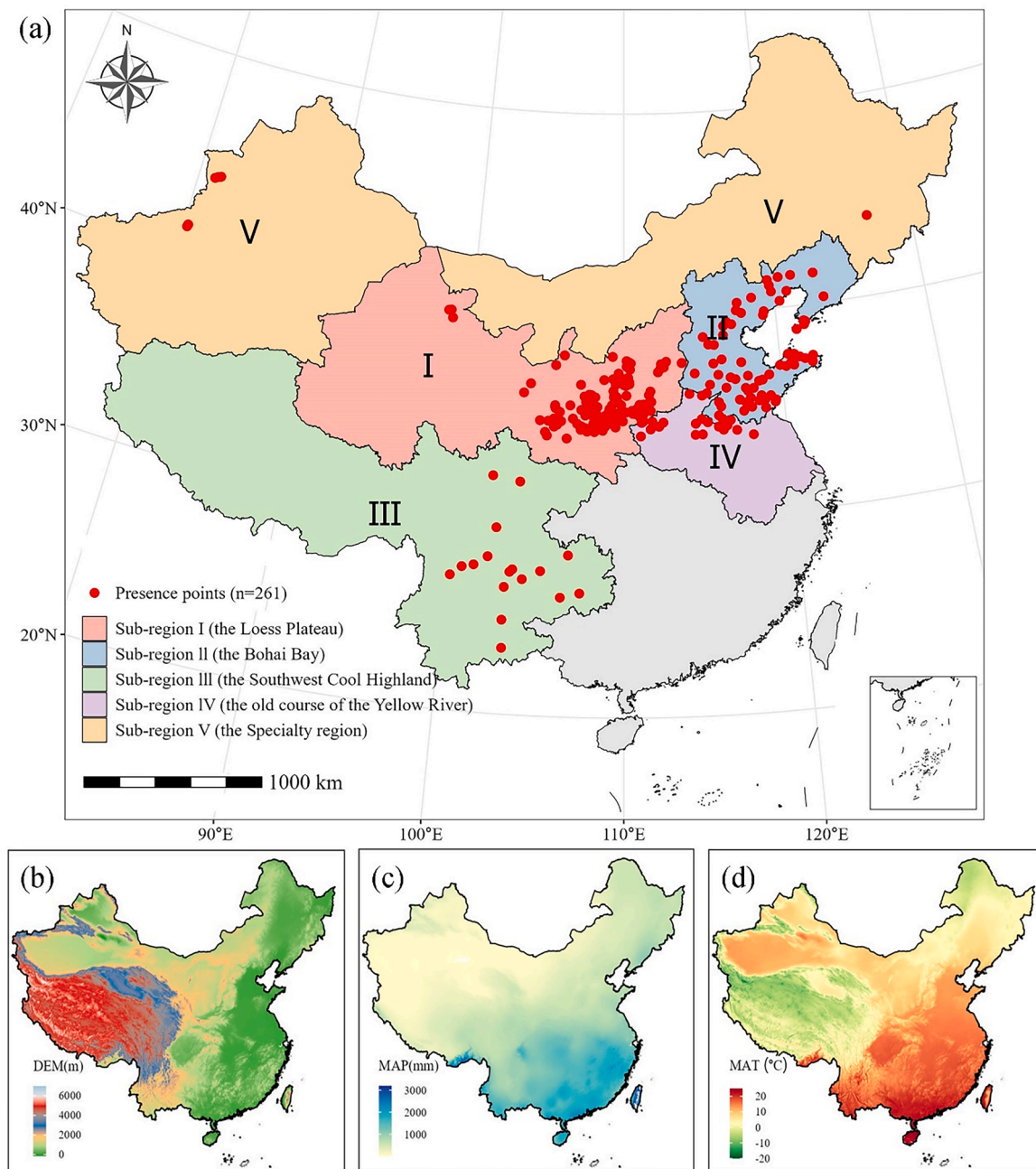


Fig. 2. The presence points of *Alternaria* leaf blotch of apple in China (a) and the environmental conditions described by the digital elevation model (DEM) (b), mean annual precipitation (MAP) from 1970 to 2000 (c) and mean annual temperature (MAT) from 1970 to 2000 (d) are shown on the bottom.

the bioclimatic data and the disease observations capture broader conditions rather than year-to-year fluctuations, ensuring that model training is based on stable and representative data, which in turn yields consistent and reliable results. Similarly, for spatial aggregation, a grid cell is marked as positive if it contains at least one recorded occurrence of ALB within its boundaries.

To reduce the impact of spatial autocorrelation during the modeling process, we used the “spThin” R package (Aiello-Lammens et al., 2015) to ensure a minimum distance of 5 km between each pair of presence points (Jiang et al., 2023; Mi et al., 2023; Xu et al., 2020). We selected a 5×5 km grid size because the spatial resolution of our raster data is 4.3 km, which is close to 5 km. After completing these processes and integrations, we obtained 261 valid ALB occurrence points (Fig. 2a).

For absence points, we generated pseudo-absence points using a random spatial sampling method (Batista et al., 2023; Ganglo, J.C., 2023;

Mi et al., 2023). Two critical questions arose: (1) Where should absence points be allocated? (2) What is the appropriate number of absence points? To determine suitable locations for absence points, we used ALB statistical data to identify regions with no occurrences of ALB from 2011 to 2019. These regions, deemed currently unsuitable for ALB, served as the basis for generating absence points (Zhang et al., 2023). Regarding the number of absence points, we followed the guideline proposed by Barbet-Massin et al. (2012), which suggests that the number of absence points should approximately equal the number of presence points to ensure the accuracy of model simulations. Therefore, we randomly selected 261 absence points using the ‘gRandom’ method in the “sdm” package in R (Barbet-Massin et al., 2012; Ganglo, 2023; Jiang et al., 2022; Naimi and Araújo, 2016).

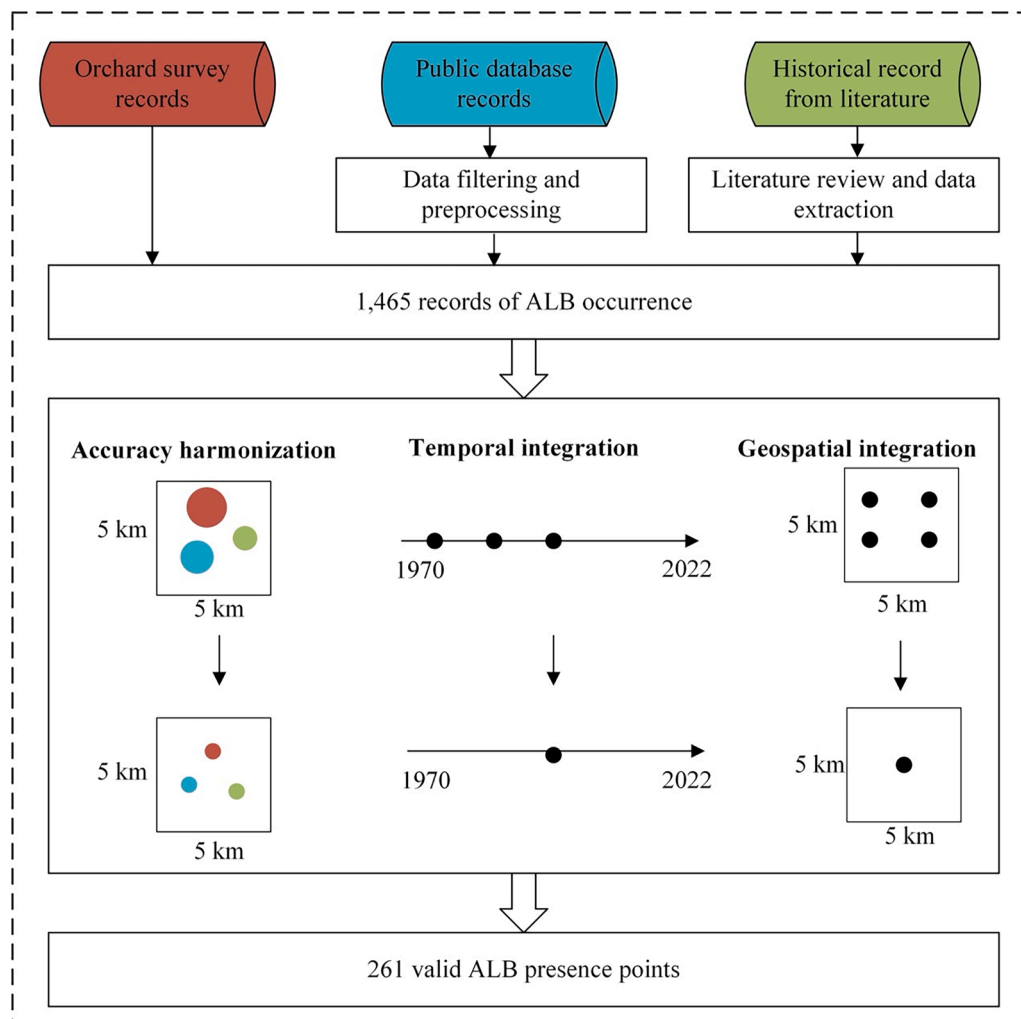


Fig. 3. The workflow for ALB presence data collection, processing, and integration.

2.4. Environmental data

2.4.1. Baseline environmental data

The period from 1970 to 2000 was selected as the baseline because it had the most extensive observational data on ALB. In this study, 23 environmental variables were utilized, including 19 bioclimatic variables (Bio1-Bio19) and four topographic variables (Elevation, Aspect, Slope, and Curvature) (Table S1). The bioclimatic variables were sourced from WorldClim (version 2.1) (<https://www.worldclim.org/>) at a resolution of 2.5 arc-minutes and were based on monthly temperature and precipitation data (Fick and Hijmans, 2017). These variables capture biological factors influencing species distribution, including annual metrics (average annual temperature and precipitation), seasonal metrics (temperature and precipitation variations across seasons), and extreme conditions (temperatures of the coldest and warmest months and precipitation during the wettest and driest quarters) (Fick and Hijmans, 2017). They are widely used in species distribution modeling (Bai et al., 2022; Booth, 2018; Booth et al., 2014; Puchaika et al., 2023).

The topographic variables were derived from a digital elevation model (DEM) and include elevation, slope, aspect, and curvature. The DEM data, originally at a 1 km resolution, were acquired from the Data Center of Resources and Environmental Sciences (<http://www.resdc.cn/>). Slope, aspect, and curvature metrics were derived using ArcGIS 10.2 software (ESRI, Redlands, CA, USA). To ensure a uniform resolution, the raster data were resampled and projected to 2.5 arc-minutes using the "raster" package in R. China's administrative boundary data

were acquired from the National Basic Geographic Information Center (<http://www.ngcc.cn/>).

2.4.2. Future climate data

Future climate data were downscaled and bias-corrected for the study sites using WorldClim v2.1 (Eyring et al., 2016). To project future climate scenarios, we used five GCMs from the CMIP6 dataset: BCC—CSM2-MR, CanESM5, CNRM-CM6-1, CNRM-ESM2-1, and MIROC6 (Table S2). These models were selected based on their extensive use in climate change studies related to species distribution in China (Shi et al., 2023; Tong et al., 2022; Zhang et al., 2018a). Their inclusion ensures that the future climate scenarios considered in this study represent key characteristics of anticipated regional climate changes, aligning with the study's goal of predicting the suitable areas of ALB under varying climate conditions. The models were evaluated under four future emission scenarios of SSPs: SSP126, SSP245, SSP370, and SSP585. The SSPs were combined with Representative Concentration Pathways (RCPs), offering various projections of future socioeconomic development and potential trends in agriculture and land use (Van Vuuren et al., 2017). The SSP126 scenario represents the lowest level of future challenges, following a pathway of historical growth with a focus on sustainability, and is projected to result in less than a 2 °C increase in temperature by 2100. The SSP245 scenario reflects intermediate challenges, maintaining historical growth patterns with reduced dependence on fossil fuels, leading to a moderate increase in temperature. The SSP370 scenario, associated with higher challenges, predicts a

temperature rise of nearly 3 °C by 2100. Finally, the SSP585 scenario, characterized by rapid socioeconomic development, increased global inequality, and a strong reliance on fossil fuels, is expected to result in a temperature increase of approximately 5 °C by 2100 (John and Post, 2022; Riahi et al., 2017; Tebaldi et al., 2021). These scenarios were analyzed for four future decades: the 2030s (average of 2021–2040), the 2050s (average of 2041–2060), the 2070s (average of 2061–2080), and the 2090s (average of 2081–2100). Considering the stability of the terrain (Xu et al., 2020; Zhang et al., 2018b), we assumed that topographic variables would remain unchanged throughout this century. To generate plausible predictions of future climate change impacts, we used these constant topographic variables in conjunction with corresponding bioclimatic variables from different periods to model the future suitable areas of ALB.

2.5. Species distribution models

To identify suitable areas for ALB under baseline and future climate change scenarios, SDMs were employed as a robust framework for linking environmental factors to ALB occurrence—BRT (Friedman, 2001), MARS (Friedman, 1991), MaxEnt (Steven et al., 2006), RF (Bradter et al., 2013), and SVM (Cortes and Vapnik, 1995)—to construct an ensemble (Ahmad et al., 2020; Araújo and New, 2007). A summary of these five SDMs can be found in Table S5. The ensemble approach ensures diversity and complementarity in addressing various data types, including non-linear relationships, missing data, and complex interactions between features (Silva et al., 2024).

Prior to training the five SDMs, we conducted a correlation analysis of 23 environmental variables to address potential multicollinearity (Fig. S4). Variables with Pearson correlation coefficients $|r| \geq 0.8$ were excluded to ensure that only those with lower intercorrelation were retained for the simulation analysis (Jayasinghe and Kumar 2019; Wang et al., 2023; Yu et al., 2022; Zhang et al., 2023). This step was crucial for improving the reliability and interpretability of the model predictions, while still maintaining a diverse set of environmental predictors,

Table 1
Environmental variables used in predicting suitable areas of ALB in China.

Type	Variables	Meaning	Units	Description
Climate Variables	Bio2	Mean diurnal range	°C	$\sum_{i=1}^{12} \frac{T_{max} - T_{min}}{12}$
	Bio3	Isothermality		$\frac{Bio2}{Bio7} \times 100$
	Bio7	Temperature annual range	°C	Bio5-Bio6
	Bio8	Mean temperature of wettest quarter	°C	
	Bio9	Mean temperature of driest quarter	°C	Mean temperature of the 3 driest months
	Bio13	Precipitation of wettest month	mm	
	Bio14	Precipitation of driest month	mm	Precipitation sums of the driest month
	Bio15	Precipitation seasonality (Coefficient of Variation)		$\frac{Std_p}{Mean_p} \times 100$
	Aspect	Aspect		SRTM elevation data
Topographic variables	Slope	Slope	degree	
	Curvature	Curvature		

Notes: T_{max} and T_{min} represent the mean monthly maximum and minimum temperatures, respectively. $\bar{p}$ indicates mean monthly precipitation. Bio5 is the maximum temperature of the warmest month, and Bio6 is the minimum temperature of the coldest month. A quarter refers to a three-month period (one-fourth of the year).

resulting in 11 retained variables (Table 1).

We allocated 70 % of the dataset for training and 30 % for testing, employing ten iterations of subsampling to account for uncertainties arising from data partitioning. Model fitting was performed using the ‘sdm’ package in R (version 4.3.3) (Naimi and Araújo, 2016), which facilitates the evaluation of predictive performance across multiple modeling methods and supports the development of ensemble models (Ganglo, J.C., 2023; Naimi and Araújo, 2016).

Model performance was assessed using the TSS (Eq. (1)) and AUC (Eq. (2)), which are widely recognized metrics for evaluating SDMs. These metrics were selected because they provide complementary evaluations of model reliability and predictive performance, particularly in SDMs studies (Barbet-Massin et al., 2012; Elith et al., 2006; Hao et al., 2020). TSS combines sensitivity (the proportion of correctly predicted presences) and specificity (the proportion of correctly predicted absences) into a single value, offering a balanced measure of model accuracy (Allouche et al., 2006). It is particularly suitable for datasets with uneven prevalence, as it is unaffected by species prevalence, making it ideal for assessing ALB’s distribution under varying geographic and climatic conditions. TSS values range from −1 to 1, with values above 0.7 generally indicating strong discriminatory power and reliable predictions (Gong et al., 2020). AUC measures the model’s ability to distinguish between presence and absence points across all possible classification thresholds (McPHERSON et al., 2004). It evaluates the overall discriminatory power of the model, independent of a specific threshold, and is robust to imbalanced datasets (Steven et al., 2006). AUC values range from 0.5 (random predictions) to 1 (perfect discrimination), with values above 0.8 considered indicative of high model reliability in predicting species distributions (Gong et al., 2020).

The combination of TSS and AUC ensures a comprehensive assessment of model performance (Hao et al., 2019; Liu et al., 2011). TSS provides threshold-dependent accuracy, while AUC evaluates threshold-independent discrimination ability. Together, these metrics enable robust evaluation of SDMs, ensuring that the models accurately capture the relationship between environmental variables and the distribution of ALB under current and future climate change scenarios. We retained models with $TSS \geq 0.7$ (Gallardo et al., 2017; Mi et al., 2023; Wang et al., 2018a) and created an ensemble model that weighted individual models by their model performance (TSS) for ALB (Jiang et al., 2022; Mi et al., 2023; Naimi and Araújo, 2016).

$$\begin{cases} TPR = \frac{TP}{TP + FN} \\ TNR = \frac{TN}{FP + TN} \\ TSS = TPR + TNR - 1 \end{cases} \quad (1)$$

where TP, FN, FP, and TN are true positive, false negative, false positive, and true negative cases predicted by the models under a given threshold, respectively. TPR and TNR are sensitivity and specificity, respectively.

$$\begin{cases} AUC = 1 - \frac{1}{m^+ \times m^-} \sum_{x^+ \in D^+} \sum_{x^- \in D^-} (w(f(x^+) - f(x^-))) \\ W(f(x^+) - f(x^-)) = 1, f(x^+) \geq f(x^-) \\ W(f(x^+) - f(x^-)) = 0, f(x^+) < f(x^-) \end{cases} \quad (2)$$

where, m^+ and m^- are sample size marked as 1 and 0, respectively; D^+ and D^- were the sets marked as 1 and 0, respectively; x^+ and x^- were the sample marked as 1 and 0, respectively; $f(x^+)$ and $f(x^-)$ were the model predicted values under a given threshold.

We assessed the relative importance of environmental variables by examining the correlation between predicted values and model outputs following variable permutation, using the ‘getVarImp’ function in the ‘sdm’ package in R (Li et al., 2023; Naimi and Araújo, 2016). The mean

relative importance of environmental variables was calculated by weighting each model according to its TSS performance. Additionally, we conducted further analysis to examine how the predicted occurrence probability of ALB changes with six key variables by studying marginal response curves. These curves depict the dependence of the probability on model prediction when other variables are held constant (Ahmadi et al., 2023; Li et al., 2020; Shi et al., 2023; Yi et al., 2018).

2.6. Projections and uncertainty analysis

The SDMs-predicted survival possibility or potential distribution of ALB ranged between 0 – 1 in each pixel. We applied the TSS-maximizing threshold method (Allouche et al., 2006; Barbet-Massin et al., 2012; Batista et al., 2023; Mi et al., 2016) to binarize the suitability of ALB into "suitable areas" and "unsuitable areas" categories, with a threshold set at 0.44. For the "suitable areas" category, we further classified the level of suitability into three levels by dividing the possibility range (0.44–1) into three equal intervals, with breakpoints at 0.63 and 0.82. Consequently, we categorized the suitability levels of ALB as follows: Unsuitable (US, 0–0.44) areas, Low suitable (LS, 0.45–0.63) areas, Moderate suitable (MS, 0.64–0.82) areas, and High suitable (HS, 0.83–1) areas. This study compares suitable areas in future scenarios with the baseline and quantifies the extent of expansion or contraction of suitable areas. Then, the suitable areas of ALB were summarized as a point, the centroid of the niche (Brown, 2014; VanDerWal et al., 2013). For each baseline and future period, the centroid of suitable areas for ALB was calculated by the following formula (Eq. (3)).

$$(x, y) = \left(\frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}, \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} \right) \quad (3)$$

where, x and y are the latitude and longitude coordinates of the centroid of the suitable areas for ALB, x_i and y_i are the geometric centroid of pixel i , and w_i is the probability of ALB occurrence of pixel i .

We analyzed various sources of uncertainty contribution using the predicted suitable area for ALB from different SDMs, GCMs, and scenarios (Scens) as an index. A three-way analysis of variance (ANOVA) was used to evaluate the uncertainty in predicting ALB's suitable area, attributed to five SDMs, five GCMs, and four Scens, including their interactions (Eq. (4)) (Jiang et al., 2022). Finally, we quantified the contribution of each component to the total uncertainty in the suitable area for ALB.

$$T_{\text{areas}} = P_{\text{areas,SDMs}} + P_{\text{areas,GCMs}} + P_{\text{areas,Scens}} + P_{\text{areas,SDMs} \times \text{Scens}} + P_{\text{areas,SDMs} \times \text{GCMs}} + P_{\text{areas,GCMs} \times \text{Scens}} + P_{\text{areas,SDMs} \times \text{GCMs} \times \text{Scens}} \quad (4)$$

where, T_{areas} is the total uncertainty in projected suitable area for ALB under future climate change scenarios; $P_{\text{areas,SDMs}}$, $P_{\text{areas,GCMs}}$, $P_{\text{areas,Scens}}$, $P_{\text{areas,SDMs} \times \text{Scens}}$, $P_{\text{areas,SDMs} \times \text{GCMs}}$, $P_{\text{areas,GCMs} \times \text{Scens}}$, and $P_{\text{areas,SDMs} \times \text{GCMs} \times \text{Scens}}$ are the proportions of uncertainty contributed by SDMs, GCMs, Scens, and their interactions, respectively.

To further support transparency and reproducibility in our research, we have shared the data and corresponding codes on GitHub at "<https://github.com/SmartAG-NWAFU/ALBClimateShift/tree/main>".

3. Results

3.1. SDMs performance

The ensemble model demonstrated robust performance, achieving an average AUC of 0.97 and an average TSS of 0.88, confirming its effectiveness in predicting the potential distribution of ALB (Fig. 4). Although the individual models also exhibited strong predictive ability, their performance varied slightly, with AUC values ranging from 0.96 to 0.97 and TSS values between 0.86 and 0.90.

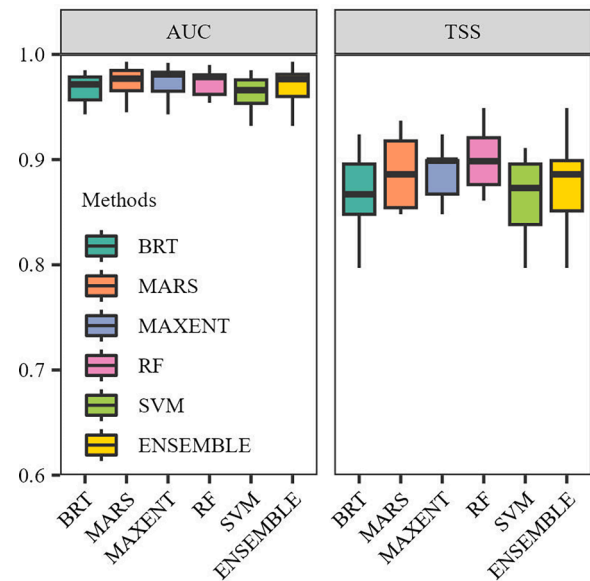


Fig. 4. The area under the receiver operating characteristic curve (AUC) and true skill statistic (TSS) values for five models and their ensemble model in predicting the suitable areas of Alternaria leaf blotch of apple: BRT (boosted regression trees), MARS (multiple adaptive regression splines), MAXNET (maximum entropy), RF (random forest), SVM (support vector machines) and ENSEMBLE (ensemble all models). Accuracy classification: AUC: 1>good>0.8>fair>0.7>poor; TSS: 1>good>0.7>fair>0.4>poor.

3.2. Contribution of environmental variables

The importance of each environmental variable in predicting the suitable areas of ALB varied significantly (Fig. 5). The mean temperature of the driest quarter (Bio9) emerged as the most influential factor, accounting for 26.96 % of the model's predictive power. This was followed by the precipitation of the driest month (Bio14), the mean temperature of the wettest quarter (Bio8), the precipitation of the wettest month (Bio13), the mean diurnal temperature range (Bio2), and the annual temperature range (Bio7). Each of the remaining variables contributed <5 % to the model. Fig. S5 illustrates the response curves for the six most

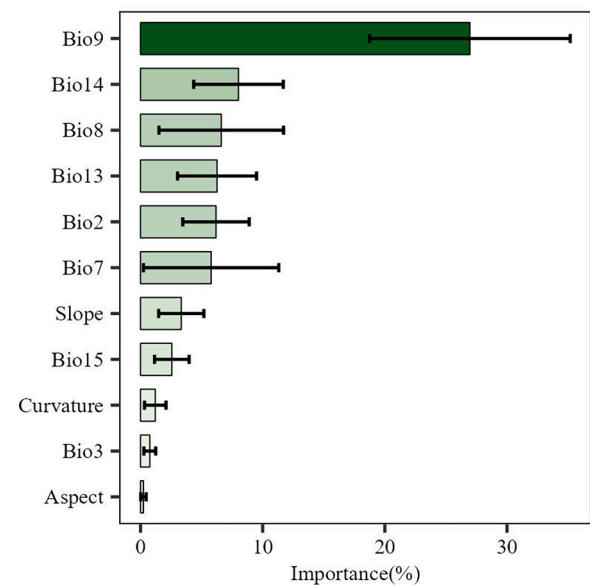


Fig. 5. Relative importance of environmental variables in predicting the suitable areas of ALB. The meaning of variables can be found in Table 1.

critical variables in relation to the probability of ALB occurrence. For Bio9, the likelihood of ALB occurrence was highest within a temperature range of -5°C to 5°C . For Bio14, the optimal precipitation range was 0–10 mm, beyond which the probability of ALB occurrence decreased sharply. The optimal temperature range for Bio8 was between 20°C and 25°C . ALB showed broad suitability for Bio13, with a higher probability of occurrence between 100 and 300 mm of precipitation. Bio2 was most favorable for ALB occurrence within a diurnal temperature range of 7.5°C to 12.5°C , with a marked decline in occurrence probability outside this range. Bio7 demonstrated a peak in ALB occurrence at temperatures between 35°C and 40°C , with reduced probability beyond this range.

3.3. The suitable areas of ALB under baseline climatic conditions

During the baseline period, the simulated distribution of ALB is presented in Fig. 6. The total suitable area (LS+MS+HS) for ALB was $1358,700\text{ km}^2$, with LS, MS, and HS area covering $473,000\text{ km}^2$ (34.81 %), $417,300\text{ km}^2$ (30.71 %), and $468,400\text{ km}^2$ (34.47 %), respectively. These suitable areas were predominantly located in the Bohai Bay, Loess Plateau, and Old Course of the Yellow River apple-planting regions. HS areas were mainly concentrated in the provinces of Gansu, Ningxia, Shaanxi, Shanxi, Henan, Hebei, Liaoning, and Shandong. These regions display a hierarchical pattern, with HS areas at the center, surrounded

by MS and LS areas. In contrast, provinces such as Xinjiang, Tibet, Sichuan, and Yunnan contained only small LS and MS areas, with no HS areas detected.

3.4. Changes in suitable areas and suitability levels of ALB under future climatic conditions

Under future climate conditions, the suitable areas and suitability levels of ALB were projected to change significantly (Fig. S6). Overall, except for the SSP126 scenario—which predicted a decrease in total suitable areas (LS+MS+HS)—the SSP245, SSP370, and SSP585 scenarios indicated expansions, primarily driven by an increase in LS areas, while MS and HS areas continued to decline (Fig. S7). Specially, when comparing the baseline period to the 2090s (Fig. 7 All-regions), the SSP126 scenario predicted an increase of $77,200\text{ km}^2$ in LS areas and $96,800\text{ km}^2$ in MS areas, while HS areas were expected to decrease by $294,800\text{ km}^2$, leading to a net reduction of $120,800\text{ km}^2$ (8.89 %) in total suitable area. In the SSP245 scenario, LS areas were projected to expand by $319,000\text{ km}^2$, and MS areas by $69,600\text{ km}^2$, while HS areas were expected to contract by $322,200\text{ km}^2$, resulting in a net increase of $66,400\text{ km}^2$ (4.89 %) in total suitable areas. The SSP370 scenario forecasted a significant increase of $654,200\text{ km}^2$ in LS areas, a minor decrease of 4300 km^2 in MS areas, and a reduction of $360,500\text{ km}^2$ in HS

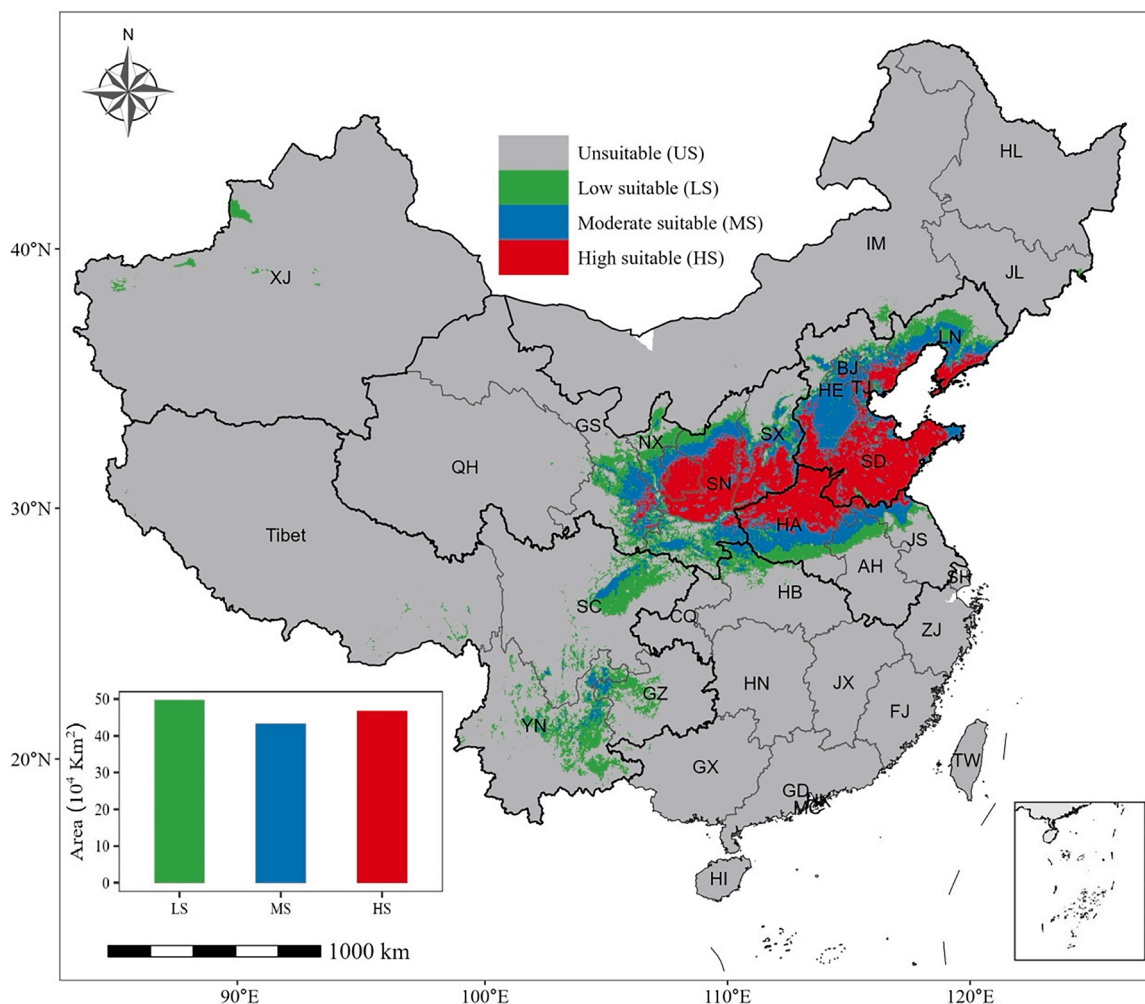


Fig. 6. Suitable areas for *Alternaria* leaf blotch of apple during the baseline period (1970–2000), categorized by suitability levels: unsuitable (US) areas, low suitable (LS) areas, moderate suitable (MS) areas, and high suitable (HS) areas. Province name and abbreviation: Guangxi: GX; Fujian: FJ; Guizhou: GZ; Yunnan: YN; Jiangxi: JX; Hunan: HN; Zhejiang: ZJ; Shanghai: SH; Chongqing: CQ; Hubei: HB; Sichuan: SC; Anhui: AH; Jiangsu: JS; Henan: HA; Tibet: Tibet; Shandong: SD; Qinghai: QH; Ningxia: NX; Shaanxi: SN; Tianjin: TJ; Shanxi: SX; Beijing: BJ; Hebei: HE; Gansu: GS; Liaoning: LN; Jilin: JL; Xinjiang: XJ; Inner Mongolia: IM; Heilongjiang: HL; Hong Kong: HK; Macao: MC; Hainan: HI; Guangdong: GD; Taiwan: TW.

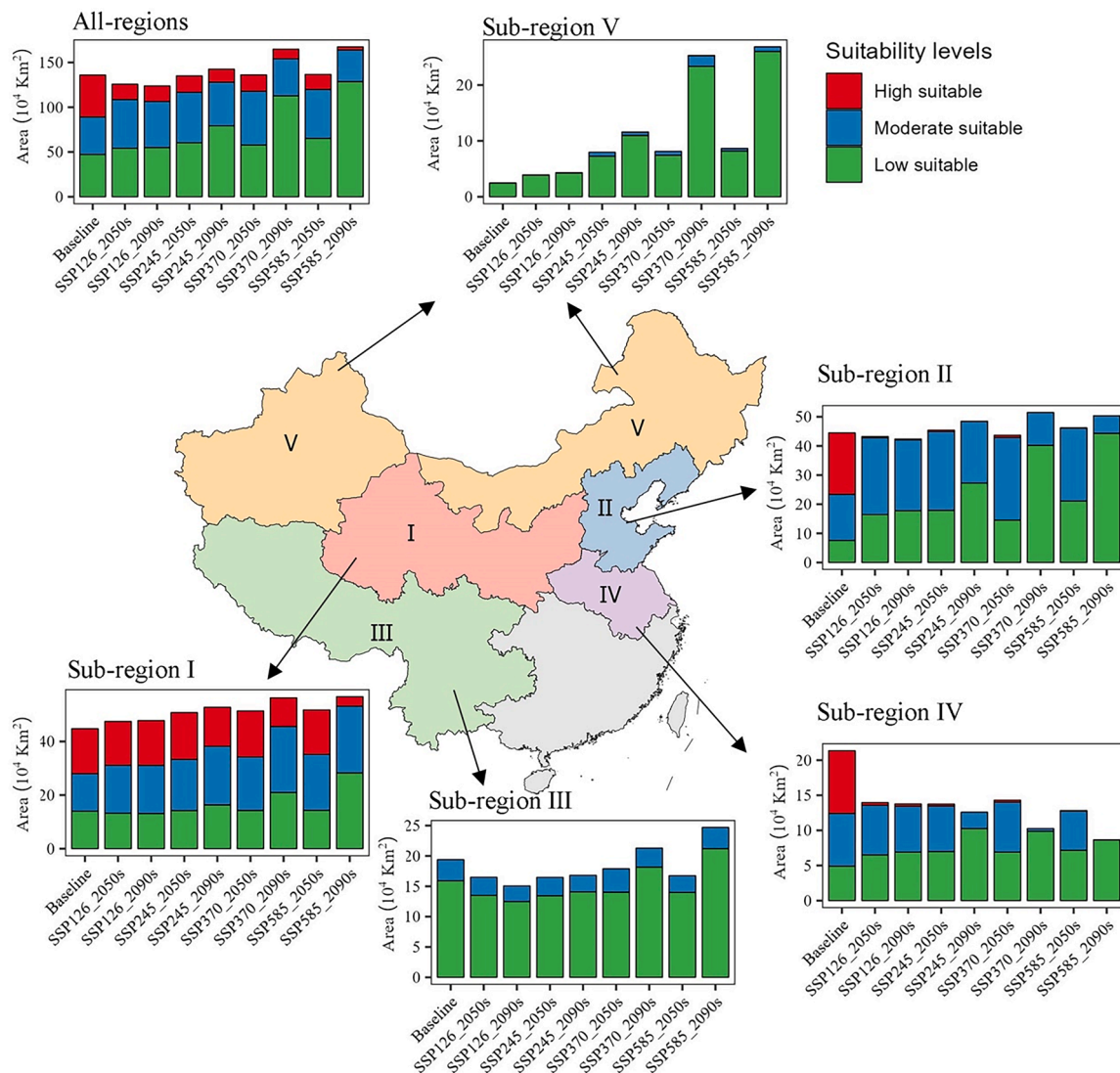


Fig. 7. Suitable areas and suitability levels of *Alternaria* leaf blotch of apple under future climatic conditions across China's apple-planting regions. I, II, III, IV, and V represent the Loess Plateau, Bohai Bay, Southwest Cold Highland area, Old Course of the Yellow River and Specialty apple-planting regions, respectively.

areas, culminating in total suitable areas increase of 289,400 km² (21.30 %). Finally, under the SSP585 scenario, LS areas are expected to grow by 813,100 km², while MS and HS areas decrease by 64,500 km² and 433,100 km², respectively, resulting in a net increase of 315,500 km² (23.22 %) in total suitable areas.

The impact of these changes varied significantly across different apple-planting regions in China (Fig. 7 Sub-region I-V). In the Loess Plateau (Sub-region I), the total suitable areas were expected to steadily increase, with MS and HS areas together comprising >50 % of the total suitable areas across all time periods. In contrast, the total suitable areas in the Bohai Bay region (Sub-region II) were expected to remain stable. In the Southwest Cold Highland region (Sub-region III), total suitable areas were expected to decrease, with only a slight increase in the 2090s under the SSP370 and SSP585 scenarios. In the Old Course of the Yellow River region (Sub-region IV), total suitable areas were projected to decrease from 213,500 km² during the baseline period to 86,400 km² by the 2090s under the SSP585 scenario. HS areas, which covered 89,300 km² during the baseline period, were expected to shrink to 3800 km² by the 2050s under SSP126 and eventually disappeared in later periods. In the Specialty region (Sub-region V), the total suitable areas were projected to increase continuously from 24,500 km² during the baseline period to 268,400 km² by the 2090s under SSP585, driven by the

expansion of LS areas.

Climate change was also expected to cause a spatial shift in suitable areas of ALB (Fig. 8 and Fig. S8). Under the low-emission scenario (SSP126), suitable areas were projected to decrease in southern regions and slightly increase in the northwest, with no significant changes across future periods. Under the SSP245, SSP370, and SSP585 scenarios, suitable areas were expected to continue expanding northwestward. As emission scenarios intensify over time, both the distance and area of expansion were projected to increase. However, regions where suitable areas decreased were expected to remain relatively stable, particularly in areas such as Sichuan, Hubei, and northern Anhui, bordering Shaanxi and southern Henan provinces.

3.5. Uncertainty sources and their contributions

The contributions of four sources of uncertainty—GCMs, SDMs, Scens, and their interactions—varied significantly over time (Fig. 9). GCMs were the largest source of uncertainty, contributing an average of 42.15 % across the four future periods. This was followed by SDMs and Scens, with average contributions of 31.50 % and 8.28 %, respectively. Among the interactions, the SDMs × GCMs interaction made the highest contribution to uncertainty, followed by Scens × GCMs, Scens × SDMs,

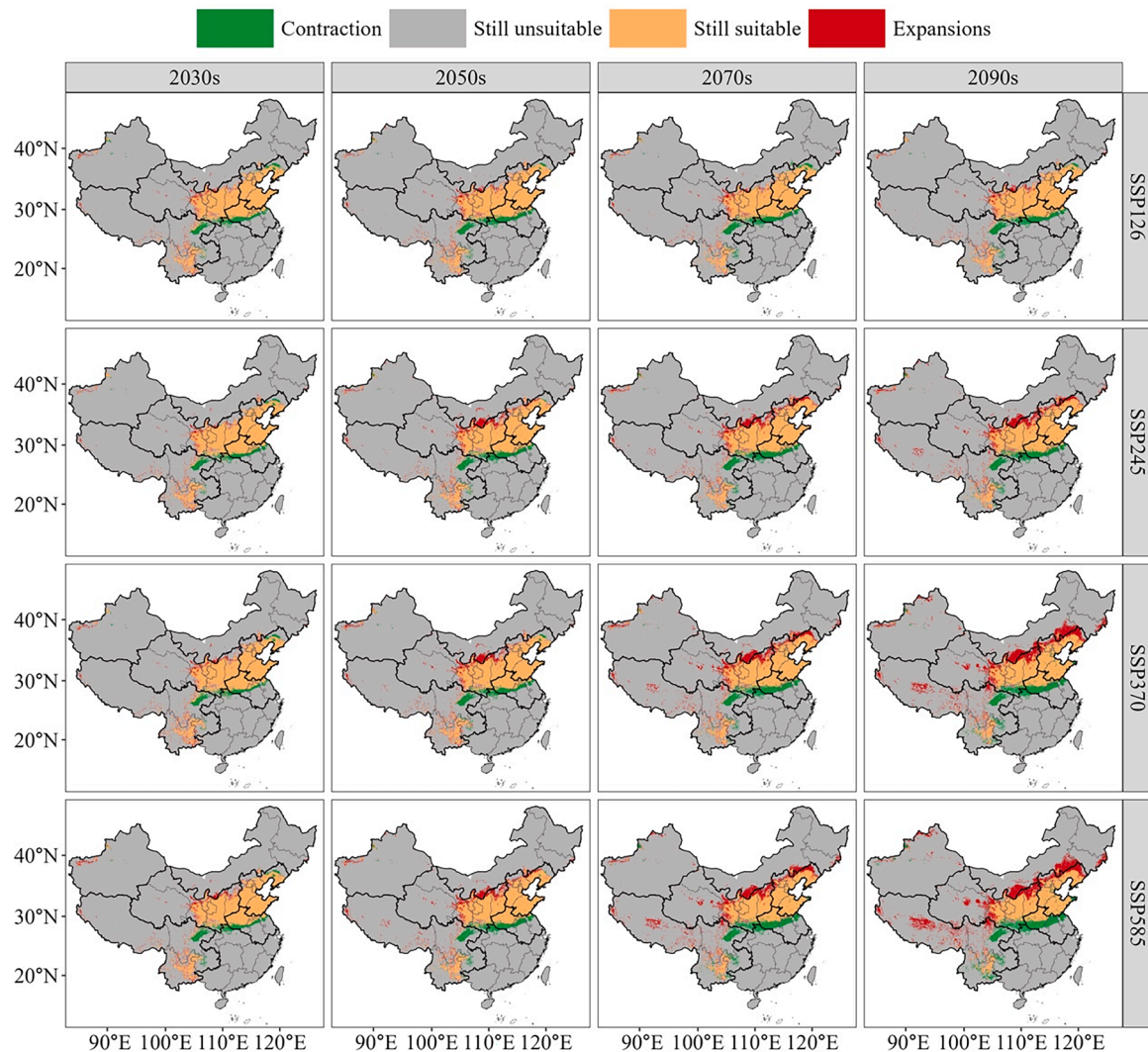


Fig. 8. Changes in the range of suitable areas for *Alternaria* leaf blotch of apple in China under future conditions, relative to the baseline period. Legend: Orange + Red – total suitable areas; Orange – persistent suitable areas; Green – areas with future range contraction; Red – areas with future range expansion; Gray – persistent unsuitable areas.

and Scens $\times$ GCMs $\times$ SDMs, with average contributions of 9.18 %, 4.22 %, 3.53 %, and 1.17 %, respectively. The contribution of uncertainty from GCMs decreased from 51.91 % in the 2030s to 29.22 % in the 2090s, while the contribution of uncertainty from Scens increased from 1.69 % in the 2030s to 17.20 % in the 2090s. The contribution of SDMs remained relatively stable at around 30 %. The SDMs $\times$ GCMs interaction was the only interaction showing a decreasing trend in uncertainty contribution, while the contributions of other interactions continued to increase over time. Overall, the analysis indicates that while GCMs remained the dominant source of uncertainty in predicting the suitable areas of ALB, the significance of scenarios increased over time. The contributions of SDMs were consistent, while the interactions among these factors varied, with certain combinations becoming more influential in later periods.

4. Discussion

This study utilized various sources of ALB occurrence data and environmental variables to model the baseline (1970–2000) and future (2030s, 2050s, 2070s, and 2090s) suitable areas of ALB in China. A three-way ANOVA was employed to assess the contributions of uncertainties from SDMs, GCMs, Scens, and their interactions to the suitable areas of ALB. The ensemble model predicted decreases in US and HS

areas of ALB for apple under SSP245, SSP370, and SSP585 scenarios, while SSP126 projected the opposite trend. Additionally, climate change caused a spatial shift in ALB suitable areas, showing a trend of expansion toward the northwest and higher elevation. Among the predicted future periods, GCMs were identified as the largest source of uncertainty, followed by SDMs and Scens. Over time, the contribution of GCMs to uncertainty decreases, while that of emission Scens increases, with the contribution of SDMs remaining relatively constant.

4.1. Impact of environmental variables on ALB suitability

Plant disease epidemics are typically governed by a complex interplay of host resistance, pathogen prevalence, environmental conditions, and intervention measures (Plant Pathology, 2005). Among these factors, environmental conditions are particularly susceptible to the effects of climate change, which can significantly alter both the occurrence and severity of plant diseases (Burdon and Thrall, 2009; Plant Pathology, 2005; Singh et al., 2023). In this study, four environmental variables—Bio9 (analogous mean winter temperature in China), Bio14 (winter precipitation), Bio8 (mean summer temperature), and Bio13 (summer precipitation)—were identified as key drivers of the spatial-temporal changes for ALB (Fig. 5). The relationships and thresholds established between these critical variables and disease

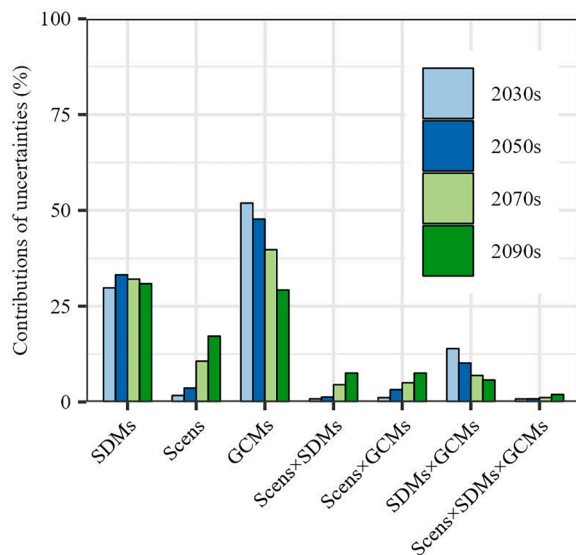


Fig. 9. Relative contributions of uncertainty components in projecting the suitable areas for *Alternaria* leaf blotch of apple in China. Uncertainty sources were broken down into global climate models (GCMs), species distribution models (SDMs), emission scenarios (Scens), and their interactions.

occurrence probability offer valuable insights into understanding the projected changes.

China's primary apple-planting regions are characterized by a monsoon climate, with concurrent heat and rain. During winter, ALB typically persists as mycelium in fallen leaves, bud scales, and diseased branches, with conidia production occurring the following spring (Wang et al., 2018b). Therefore, winter average temperature (Bio9) and precipitation (Bio14) directly influence the overwintering of AMR spores and their production in early spring. Additionally, low winter temperatures can cause frost damage to apple trees, exacerbating disease occurrence in the following year (Xu et al., 2020). During summer, when apple trees are in full foliage and orchards are relatively enclosed, there is a peak in ALB incidence (Wang et al., 2018b). Hu et al. (2005) noted that summer precipitation was the dominant factor in ALB outbreaks, with nearly every rainfall event triggering a peak in disease incidence. The amount of summer rainfall and air humidity levels are essential conditions for widespread ALB infection (Harteveld et al., 2014; Hu et al., 2006), explaining why summer rainfall (Bio13) and summer temperature (Bio8) play significant roles.

Besides bioclimatic and topographic variables, other environmental factors, such as soil properties, may also influence the distribution of ALB. For example, Cao (2018) found that ALB occurrence was related to calcium (Ca) content, with calcium deficiency increasing susceptibility to the disease. Similarly, increasing potassium (K) and phosphorus (P) within normal ranges reduced plant vulnerability. Changes in soil properties could also enhance crop and disease resilience to climate change (Qiao et al., 2022). Soil factors were not included in this study due to unclear mechanisms by which soil affects the occurrence and development of ALB. Furthermore, air humidity was excluded from the bioclimatic variables because precipitation-related variables were thought to partially account for it (Bai et al., 2022).

This study primarily focuses on environmental variables; however, it is crucial to recognize the potential influence of additional factors, including orchard management practices, pathogen transmission dynamics, and host susceptibility. Management interventions such as fungicide application, pruning, and irrigation schedules can significantly alter orchard microclimates and subsequently affect disease dynamics (Tongle and Shutong, 2006; Wang, 2024; Zaller et al., 2023). In addition, AMR is a multi-host pathogen that affects species such as pear trees and other woody plants (Ebadi and Ebadi, 2024; Kang et al., 2013).

The spatial distribution and density of these alternative hosts could impact ALB prevalence and spread, especially in regions experiencing increased pear cultivation. Another important consideration is the variability in apple cultivar susceptibility across different regions. However, these factors were not included due to limited availability of standardized data on orchard management practices, alternative host distributions, and apple cultivar susceptibility. Future research should focus on data collection and standardization to better understand their impacts on ALB dynamics.

Recent studies indicate that numerous pathogens have undergone significant strain changes in response to climate change and evolving agricultural practices (Rai, 2009). In addition, Garrett et al. (2006) also mentioned that the adaptive potential of plant and pathogen populations may be an important factor in predicting the extent of climate change impacts. These shifts have led to variations in the adaptability and virulence of different strains across various regions. Climate changes, including alterations in temperature and humidity, have driven these strains to mutate, potentially increasing their aggressiveness (Biota, 2002; Rudman et al., 2022). However, current data is insufficient to train models that accurately study strain adaptation (Kelly et al., 2012). Consequently, for the purposes of this study, we assume that the environmental dependency conditions of antimicrobial resistance will remain relatively stable throughout this century.

4.2. Projected shifts in the suitable areas for ALB

In the future, except for the SSP126 scenario, which predicted a decrease in total suitable areas (LS+MS+HS) for ALB, the other emission scenarios indicated increases in total suitable areas. This increase was primarily driven by the expansion of low suitable areas, while high suitable areas continued to decline (Fig. S7). Under high emission scenarios, significant temperature increases could convert previously unsuitable areas (cold northern regions) into low suitable areas (Fig. 8). Conversely, when the temperature increases beyond the optimal range for AMR survival, this warming will not favor the development of ALB. Consequently, HS areas in the central region transitioned into MS and LS areas (Fig. 6 and Fig. S6).

Previous studies have shown that climate change often drives species to shift towards higher latitudes and elevations (Chen et al., 2011; Couet et al., 2022; Parmesan and Yohe, 2003). In our study, we found that under future climate conditions, the suitable areas for ALB were projected to move towards northwestern and higher elevations (Fig. 8 and Fig. 10). As emission scenarios intensify, the cumulative distance and elevation of this expansion were expected to increase. By the 2090s, the suitable areas of ALB were projected to shift 137 km under SSP126, 171 km under SSP245, 229 km under SSP370, and 263 km under SSP585 (Fig. 10a). Similarly, the shift in elevation was expected to be 288 m under SSP126, 576 m under SSP245, 617 m under SSP370, and 680 m under SSP585 (Fig. 10b). Our findings align with previous studies that describe pathogen shifts to higher elevations (Bebber, 2015; Porta et al., 2008; Tang et al., 2021). However, our study also reveals a northwestward shift, rather than a strictly northward movement to higher latitudes. This deviation is likely due to China's unique monsoon climate, where climate change is resulting in warmer and more humid conditions in the northwest (Ding et al., 2022), creating a favorable environment for the survival of the ALB pathogen.

4.3. Impact of ALB suitable areas changes on future apple cultivation management

This study reveals a critical finding that ALB is projected to shift towards higher elevations and northwestern regions due to climate change. By employing multi-scenario modeling under CMIP6 data, we are able to forecast how different climate pathways will impact the potential distribution of ALB. This approach offers a technical advantage by accounting for multiple potential climate futures, which enhances the

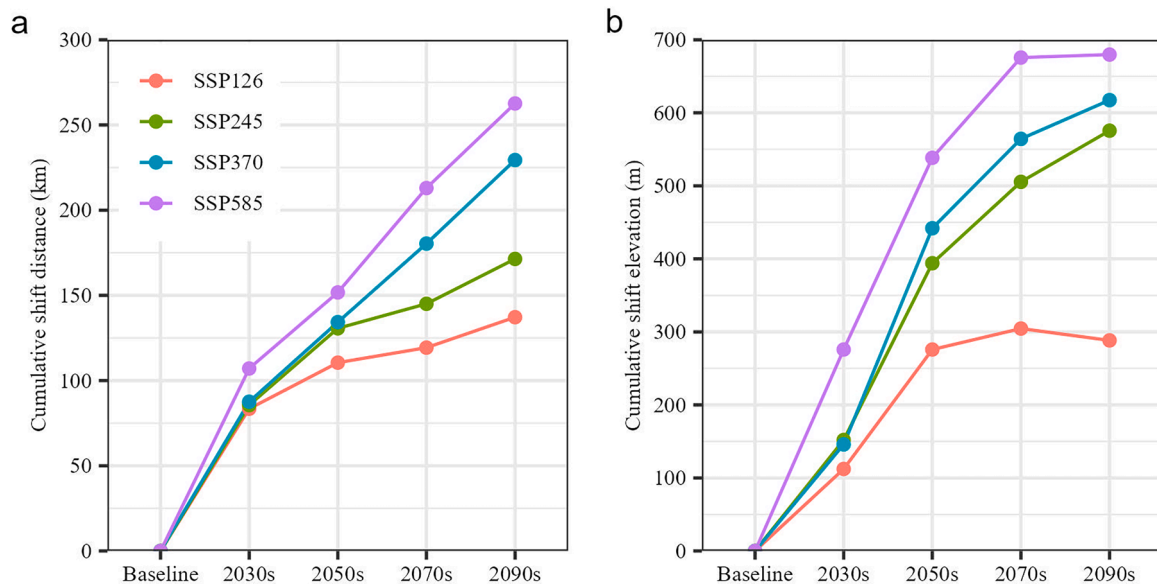


Fig. 10. Cumulative shift distance and elevation of the centroid of suitable areas for *Alternaria* leaf blotch of apple in China under four scenarios (SSP126, SSP245, SSP370, and SSP585) for four future periods (2030s, 2050s, 2070s, and 2090s).

reliability of the projections and allows for more informed decision-making in disease management (López-Pujol et al., 2006; Zhao et al., 2022).

The shift in ALB's suitable areas to higher elevations and north-western regions has significant practical implications for apple growers. For regions where ALB suitability is projected to increase—particularly in higher elevations—growers may need to adapt by relocating orchards or altering cultivation practices. In these areas, heightened disease monitoring, early detection, and targeted interventions (such as using fungicides or resistant apple varieties) will be essential to prevent crop losses (Cabrefiga et al., 2023; Ramírez-Gil et al., 2021). As ALB moves to higher altitudes, apple cultivars may need to be re-evaluated for their resistance to the disease in these new environments. In contrast, for regions where ALB suitability is expected to decline, especially in lower elevation areas, growers may experience reduced disease pressure, but could face other challenges such as altered growing seasons, water scarcity, or new pest threats (Savary and Willocquet, 2020; Singh et al., 2023).

The long-term implications of these shifts necessitate forward-looking planning. By integrating climate projections into management strategies, apple growers can better prepare for ALB's spatial shifting. This could involve selecting more resistant apple varieties, improving irrigation efficiency, and adjusting orchard management practices. As Lamichhane et al. (2015) suggested, policy, research, and extension should adopt a “no regrets” strategy to prepare for worst-case scenarios, enhancing the resilience of cropping systems and identifying and strengthening adaptive capacity.

4.4. Assessing the uncertainty in ALB suitable areas projections

Typically, the uncertainty in SDMs predictions arises from multiple sources, such as the SDMs, GCMs, scenarios, data collection bias, selection of environmental variables, and model parameters (Brodie et al., 2022; Buisson et al., 2010; Watling et al., 2015). Among these, previous studies have identified SDMs, GCMs, and scenarios as the most common and significant contributors to uncertainty (Beaumont et al., 2007; Cheung et al., 2016; Goberville et al., 2015; Morley et al., 2020). Consequently, this study focuses on analyzing the impact of these factors on prediction outcomes.

Our results revealed that GCMs were the primary source of uncertainty in predicting suitable areas for ALB, contributing an average of

42.2 % to the overall uncertainty, followed by SDMs at 31.5 % and scenarios at 8.28 % (Fig. 9). However, Thuiller et al. (2019) identified SDMs and RCPs (scenarios in CMIP5) as the most significant contributors to uncertainty in projections, with GCMs playing a lesser but still important role. Wiens et al. (2009) and Beale and Lennon (2012) emphasized that model structure (analogous to SDMs) and data quality were major sources of uncertainty, with the influence of GCMs varying depending on the study context. The differences between our findings and those of other studies can be attributed to several factors. One possible explanation is that the unique ecological and environmental characteristics of ALB make its distribution particularly sensitive to the climatic variables projected by GCMs. Additionally, the varying scales of the study area and GCM models may also play a role. This study covers all apple-planting regions across China, which introduces significant climatic variability due to the large regional differences in climate (as shown in Fig. 2b, c, and d). This study also found that the contribution of GCMs to uncertainty decreased over time—from 51.9 % in the 2030s to 29.2 % by the 2090s—while the contribution of scenarios increased from 1.69 % in the 2030s to 17.2 % by the 2090s (Fig. 9). This trend is consistent with the findings of Goberville et al. (2015), which showed that GCMs' contributions exceeded those of scenarios in the near term but converge towards the end of the century.

Furthermore, this study has innovatively revealed that the interaction between SDMs × GCMs contributed significantly to the uncertainty in prediction results, accounting for 9.18 % of the total uncertainty. This contribution was much higher than that of other combinations, such as Scens × GCMs (4.22 %), Scens × SDMs (3.53 %), and Scens × SDMs × GCMs (1.17 %) (Fig. 9). This finding suggests that the interaction between SDMs and GCMs is a significant source of uncertainty, especially when dealing with complex ecosystems and climate change predictions. Given this, accounting for the interactive effects of GCMs and SDMs allows for a more comprehensive assessment of the uncertainties in predicting suitable areas for ALB.

It is important to note that, while SDMs, GCMs, and scenarios are identified as the primary sources of uncertainty in our study, other potential sources—such as data collection bias, selection of environmental variables, and model parameters—should not be overlooked. Firstly, data collection bias may impact the occurrence data for ALB. If the data exhibits regional bias, the model predictions may lack representativeness, which would affect the accuracy of suitable area predictions (Arenas-Castro et al., 2022; Watling et al., 2015). Furthermore, the

choice of environmental variables is another significant source of uncertainty. While we selected key climatic factors, such as temperature and precipitation, other environmental factors that may influence ALB distribution were not considered. The variables overlooked in the modeling process can introduce new uncertainties into the prediction results (Bai et al., 2022; Ganglo, J.C., 2023; Xu et al., 2020).

Additionally, due to regional differences in pathogen prevalence and fungicide resistance, tailoring model parameters to specific regions can enhance predictive accuracy (Anderson and Gonzalez Jr, 2011; Luna et al., 2024). Such variations are influenced by local management practices, which affect both disease intensity and frequency. Furthermore, different fungicide usage patterns can lead to varying levels of resistance. Incorporating these region-specific factors into the model would likely yield more accurate predictions by reflecting the distinct ecological and management contexts of each region.

4.5. Limitations and future work

Several limitations of this study should be noted, which may require further refinement. Firstly, this study utilizes three sources of ALB occurrence data, which provide comprehensive spatial coverage across most of China's apple-planting regions. However, the temporal coverage remains limited, as long-term, continuous records of ALB occurrences are challenging to obtain. Machine learning models, including the SDMs employed in this study, generally perform well for interpolation within the observed data range. However, their predictive accuracy may decrease when extrapolated to future conditions beyond the scope of the training data (Muckley et al., 2023; Zhan et al., 2022). To address these uncertainties in future research, integrating machine learning approaches with mechanistic models presents a promising pathway (Zhao et al., 2024). Mechanistic models, which incorporate biological and ecological processes such as disease dynamics, pathogen-host interactions, and environmental thresholds, can complement statistical models by offering deeper insights into the fundamental drivers of ALB distribution (Rossi et al., 2008). Hybrid models that combine statistical and mechanistic frameworks have demonstrated improved reliability in forecasting plant diseases, particularly for long-term projections under evolving climate conditions (González-Domínguez et al., 2023; Saubin et al., 2024).

Secondly, this study assumes that topographic factors, such as slope, elevation, and aspect, remain stable in the future. However, land use changes and shifts in vegetation cover are likely to alter these factors, potentially affecting the suitability of areas for ALB. Due to data limitations and the need for model simplification, we have assumed static topography. In future studies, where data permits, dynamic topographic and land use changes should be incorporated to better account for their potential impact on ALB suitability.

Finally, this study's uncertainty analysis considers only three major sources—SDMs, GCMs, and Scens—while excluding other factors such as model parameters, the completeness of survey data, and the number of predictor variables (Brodie et al., 2022; Buisson et al., 2010; Watling et al., 2015). Future research should expand the scope of uncertainty analyses by incorporating a broader range of factors, such as model parameter sensitivity and the potential influence of incomplete or biased survey data. Additionally, it will be crucial to explore the impact of varying predictor variables and interactions among them, as this may further refine model accuracy and reduce uncertainty in predicting species distributions under future climate scenarios.

5. Conclusion

This study integrated SDMs with multi-source environmental data, including climatic and topographic factors, to simulate the suitable areas of ALB on apple in China. The five SDMs effectively predicted both the suitable areas and suitability levels of ALB. The mean temperature of the driest quarter and the precipitation of the driest month were

identified as the most critical environmental variables influencing ALB's suitability. Under the baseline climate conditions, the simulations indicated that ALB's suitable areas were primarily concentrated in the Bohai Bay, the Loess Plateau, and the Old Course of the Yellow River apple-planting regions. For the future, except for the SSP126 scenario, which predicted a decrease in total suitable areas for ALB, the SSP245, SSP370, and SSP585 scenarios indicated increases in total suitable areas, primarily driven by the expansion of low suitable areas, while moderate and high suitable areas continued to decline. Additionally, the suitable areas of ALB were expected to shift towards northwestern and higher elevations. The northern Bohai Bay and Loess Plateau regions are likely to become key areas for ALB prevention and control in the future. Moreover, the study found that GCMs were the largest source of uncertainty, contributing an average of 42.15 %, followed by SDMs and Scens, with contributions of 31.50 % and 8.28 %, respectively. By identifying future hotspots and suitable areas, this study provides valuable insights for safeguarding apple production against the impacts of climate change. Additionally, these findings underscore the complexities of predicting disease suitable areas under climate change and highlight the importance of using multiple models and scenarios for accurate forecasting.

CRediT authorship contribution statement

Bin Chen: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Gang Zhao:** Writing – original draft, Supervision, Formal analysis, Conceptualization. **Qi Tian:** Investigation, Formal analysis. **Linjia Yao:** Investigation. **Genghong Wu:** Writing – review & editing. **Jing Wang:** Writing – review & editing. **Qiang Yu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.agrformet.2025.110464](https://doi.org/10.1016/j.agrformet.2025.110464).

Data availability

Data will be made available on request.

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